Topology-Constrained Failure Attribution: Physically Feasible Counterfactual Explanations for GNN-Based Traffic Forecasting

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Abstract. This paper presents a novel research paper at the intersection of explainable ML on graphs and infrastructure network analysis. Graph Neural Networks (GNNs) are increasingly deployed for failure prediction in backbone networks, yet existing counterfactual explanation methods—including CF-GNNExplainer and GCFExplainer—impose no physical constraints on recommended interventions: a perturbation may disconnect the network, saturate link capacities, or violate routing feasibility, rendering it operationally useless. We propose **Topology-Constrained Failure Attribution (TCFA)**, the first counterfactual explanation framework for GNN-based traffic forecasters to jointly enforce connectivity (C1), capacity (C2), and routing feasibility (C3), without assuming ground-truth knowledge of which edges have failed. Candidates are ranked by a score that multiplicatively couples physical feasibility with predicted failure-probability reduction. We further introduce Fix@k, a new actionability metric measuring whether applying the top-k ranked interventions resolves the predicted failure — a criterion orthogonal to Hit@k and MRR that we argue should accompany standard retrieval metrics wherever GNN explanations guide real-world interventions. Evaluated on Abilene and GÉANT across five independent failure configurations each, TCFA achieves Hit@1 = 100% and MRR = 1.000 on both datasets, and Fix@3 = 94% / 100% respectively—versus **0%** for an unconstrained ablation. A 74.7% constraint violation rate among unconstrained candidates confirms that physical feasibility is not a post-hoc filter but the primary driver of attribution performance.

Keywords: Graph neural networks · Counterfactual explanation · Network failure attribution · Topology constraints · Traffic forecasting · Explainability · Backbone networks · Actionability metrics

1 Introduction

Wide-area backbone networks underpin global Internet traffic, and even brief link failures cause measurable economic and quality-of-service consequences [12]. As

GNNs demonstrate strong performance on backbone traffic forecasting [9, 20], their deployment in operational settings raises a new problem: when these models predict an imminent failure, the prediction itself offers no guidance on *which* links to inspect or restore first. The question is not merely interpretive—it is time-critical. An operator who must choose among dozens of candidate interventions in real time needs a ranked list they can act on, not a gradient heatmap they must translate into action themselves.

Counterfactual explanation methods address exactly this need in principle: they identify the smallest change to the input that would flip the model's prediction, yielding an explanation of the form "the model would not have predicted failure if edge (u, v) had been operational." CF-GNNExplainer [11] formalises this for arbitrary GNN classifiers, and GCFExplainer [6] extends it to group-level counterfactuals. However, both methods treat the graph perturbation space as unconstrained: any subset of edges may be added or removed, regardless of whether the resulting graph is physically realisable. Applied to a network topology, such methods may recommend restoring a link whose reinstatement would force all inter-domain traffic onto a single remaining path, saturating its capacity, or suggest adding an edge between two nodes with no physical fibre. These explanations are mathematically valid counterfactuals but are operationally actionable in name only.

The gap between physical feasibility and counterfactual graph explanation has been noted but never closed for infrastructure networks. GCFExplainer [6] observes that domain constraints can in principle be incorporated by pruning the candidate graph space, but provides no instantiation of this idea for any network-systems domain. CF-GNNExplainer [11] likewise provides no mechanism for constraint enforcement. To our knowledge, no prior work has embedded physical network constraints—connectivity, link capacity, and routing feasibility—into a counterfactual GNN explanation framework for traffic networks.

We address this gap with **Topology-Constrained Failure Attribution (TCFA)**, a counterfactual explanation framework that integrates physical network constraints directly into the perturbation search. TCFA restricts candidate interventions to *edge restorations*—links present in the original healthy topology but absent in the current failure snapshot—ensuring that every candidate corresponds to a plausible operator action. Each candidate is evaluated against three constraints: (C1) *connectivity*, requiring the post-intervention topology to remain connected; (C2) *capacity*, requiring that rerouted traffic does not exceed per-link capacity bounds derived from historical load; and (C3) *routing feasibility*, requiring all origin–destination flows to remain routable under shortest-path routing [4] on the modified topology. Candidates satisfying all three constraints are ranked by a composite score that multiplicatively couples feasibility with the reduction in predicted failure probability. An *unconstrained* variant ($\phi \equiv 1$) serves as a CF-GNNExplainer-style proxy baseline, isolating the contribution of the physical constraints from the model-based scoring.

Contributions.

1. **Problem formulation.** We formalise the *topology-constrained counterfactual failure attribution* problem (Definition 1) for GNN-based failure classifiers, embedding connectivity, capacity, and routing feasibility as explicit constraints on the counterfactual search space—the first such formulation for network infrastructure explanation.
2. **The TCFA framework.** We propose a constrained search algorithm that (a) verifies each candidate edge restoration against all three physical constraints using the observed origin–destination traffic matrix and a data-driven capacity estimate (P_{95}(historical load) $\times$ 1.5), and (b) ranks interventions by the composite score $\phi(\delta) \cdot \max(0,\, \hat{P}_{\text{base}} - \hat{P}_{\delta})$, where $\phi(\delta) \in [0, 1]$ is a feasibility weight that degrades multiplicatively under capacity violation and collapses to zero for connectivity or routing failures.
3. **Fix@k: a new actionability metric.** We introduce Fix@k (Definition 2), which measures the fraction of failure samples for which applying the top-k TCFA-ranked interventions reduces predicted failure probability by at least $\delta_{\text{abs}} = 0.05$. We additionally report a model-relative variant (Fix@k_{rel}, $\delta_{\text{rel}} = 0.40$) that normalises for classifier calibration differences across datasets.
4. **Empirical evaluation on two backbone networks.** We evaluate TCFA on Abilene (12 nodes, 15 edges) and GÉANT (23 nodes, 38 edges) using empirical traffic matrices, evaluating on five held-out failure seeds $\{7, 13, 21, 55, 99\}$. TCFA achieves Hit@1 = 100% and MRR = 1.000 on both networks across all seeds, Fix@3 = 94% on Abilene and Fix@3 = 100% on GÉANT, versus 0% for the unconstrained ablation. A mean constraint violation rate of 74.7% on Abilene confirms that feasibility filtering is active and consequential.

The remainder of this paper is organised as follows. Section 2 reviews related work. Section 3 formalises the problem and describes TCFA. Section 4 presents experimental results. Section 5 discusses findings and limitations. Section 6 concludes.

2 Related Work

2.1 Graph Neural Networks for Traffic Forecasting

Traffic forecasting over backbone networks is fundamentally a graph problem because links and routers form a natural relational structure and methods that respect this tend to work better. DCRNN [9] captures this by treating flow as something that diffuses across edges, using graph convolution alongside recurrent units to track how patterns evolve in both space and time. STGCN [22] arrives at similar accuracy but drops the recurrence entirely, relying on stacked convolutional blocks that are faster to train. Graph WaveNet [20] goes furthest - rather than assuming you know the connectivity upfront, it learns which nodes are meaningfully related directly from observed traffic.

2.2 GNN Explainability

Making sense of what a GNN actually learned is harder than it sounds. Some methods go after edge or feature masks, trying to isolate which parts of the input drove the prediction [21, 10]. Others take a generative angle, producing a spread of counterfactual examples to show what the input could have looked like for a different outcome [2, 3]. Both families are useful, but they share a common blind spot, they treat the space of possible perturbations as unconstrained, which works fine on benchmark graphs but falls apart when the graph represents real physical infrastructure.

2.3 Root Cause Analysis in Network Systems

Root cause analysis in networked systems has traditionally drawn on alarm correlation [12], rule-based fault localisation, and causal inference over dependency graphs [13]. GNN-based anomaly detection has more recently improved sensitivity to spatially correlated faults, though these methods largely stop at detection — they identify that something is wrong without addressing which intervention would actually resolve it. This distinction matters in practice. TCFA targets the attribution step directly, combining GNN-based failure prediction with counterfactual interventions that are anchored in the physical constraints of the network rather than treating the topology as an abstract graph.

2.4 Constrained Counterfactual Explanation

Domain constraints in counterfactual explanation have received the most attention in algorithmic recourse, where inputs are tabular and feasibility reduces to bounds on individual features [19, 17]. DICE [14] generates counterfactuals that respect user-specified feature ranges, while FACE [15] keeps suggested changes realistic by restricting the counterfactual path to high-density regions of the data. Both are sensible approaches for tabular settings, but neither carries over to graph topology interventions. The reason is that feasibility for a network graph is not about feature ranges, it is about whether the graph remains connected, whether all origin-destination pairs can still be routed, and whether rerouted traffic stays within link capacity. GCFExplainer [6] acknowledges that structural constraints could in principle be used to prune the counterfactual search space, but does not develop this idea for any concrete network domain. TCFA fills that gap, enforcing connectivity, capacity, and routing feasibility directly as hard constraints within a unified counterfactual scoring framework for wide-area backbone networks.

3 Methodology

3.1 Problem Formulation

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a backbone network topology with node set $\mathcal{V}$ and edge set $\mathcal{E}$. At each timestep t, a per-link traffic vector $\mathbf{X}_t \in \mathbb{R}^{|\mathcal{E}|}$ records the observed load on each edge, and an origin–destination matrix $\mathbf{T}_t \in \mathbb{R}^{|\mathcal{V}|\times|\mathcal{V}|}$ captures demand between node pairs. A GNN classifier $f : \mathcal{G}_t \to [0, 1]$ outputs the probability of a network failure at timestep t, where $\mathcal{G}_t$ is the graph snapshot augmented with node features derived from $\mathbf{X}_t$. When a failure occurs, some subset of edges $\mathcal{E}_f \subset \mathcal{E}$ disappears from the topology, leaving a degraded graph $\mathcal{G}_t^f = (\mathcal{V}, \mathcal{E} \setminus \mathcal{E}_f)$. We define $\Delta = \{\delta \subseteq \mathcal{E} \mid \delta \cap (\mathcal{E} \setminus \mathcal{E}_f) = \emptyset\}$ as the set of candidate edge restorations—those edges currently absent from the failure snapshot that could plausibly be reinstated.

Definition 1 (Topology-Constrained Counterfactual Failure Attribution). *Given a GNN classifier $f : \mathcal{G}_t^f \to [0,1]$ and a failure snapshot $\mathcal{G}_t^f$ where $f(\mathcal{G}_t^f) > \tau$, the topology-constrained counterfactual is:*

$$\delta^* = \arg\min_{\delta \in \Delta} f(\mathcal{G}_t^f \oplus \delta) \tag{1}$$

subject to (C1) $\mathcal{G}_t^f \oplus \delta$ remains connected, (C2) rerouted traffic does not exceed the per-link capacity bounds $c_e = 1.5 \times \mathrm{P}_{95}(\mathrm{traffic}_e)$, and (C3) every origin–destination flow stays routable under shortest-path routing on $\mathcal{G}_t^f \oplus \delta$.

3.2 Node Feature Construction

Each graph snapshot is represented using node features that are derived purely from topology and traffic observations, with no leakage of future information. For node v at timestep t, the feature vector $\mathbf{x}_v^t \in \mathbb{R}^5$ is constructed as:

$$\mathbf{x}_v^t = \left[\deg(v),\ b(v),\ \sum_{e \ni v} x_{t,e},\ \sum_{e \ni v} (x_{t,e} - x_{t-1,e}),\ \frac{\sum_{e \ni v} |x_{t,e}|}{\sum_{e'} |x_{t,e'}| + \epsilon}\right], \tag{2}$$

where $\deg(v)$ is the degree of node v, $b(v)$ is its betweenness centrality, and $x_{t,e}$ is the normalised traffic observed on edge e at time t.

Structural features are recomputed at each timestep to reflect whatever topology is currently active rather than the original healthy graph. For GÉANT, the feature dimension is extended to 10 by additionally incorporating clustering coefficient, PageRank, and multi-channel edge attributes, accounting for the larger and more heterogeneous topology.

3.3 GNN Failure Classifier

The classifier f is selected from GCN [8], GAT [18], and GraphSAGE [5] based on validation F1-score. Each architecture uses three message-passing layers with hidden dimension 64, dropout (0.2), and global mean pooling, followed by a two-class log-softmax output. The predicted failure probability is extracted as $\hat{p}_t = \exp(f(\mathcal{G}_t^f))[1]$, where $[\cdot][1]$ indexes the failure class and $\exp(\cdot)$ recovers the probability from the log-probability. Class imbalance is addressed through inverse-frequency weighting.

3.4 TCFA Constrained Search Algorithm

TCFA evaluates each candidate edge restoration $\delta = (u, v) \in \mathcal{E}$ across the full original edge set by (i) constructing the post-intervention graph $\mathcal{G}' = \mathcal{G}_t^f \cup \{(u, v)\}$, (ii) computing the feasibility score $\phi(\delta)$, and (iii) querying the classifier on the updated topology. The composite constraint satisfaction score is:

$$\phi(\delta) = \chi_{C1}(\delta) \cdot \chi_{C2}(\delta) \cdot \chi_{C3}(\delta), \tag{3}$$

where $\chi_{C1}(\delta) \in \{0, 1\}$ indicates connectivity preservation (hard constraint), $\chi_{C2}(\delta) = \max(0, 1 - \rho_{\max})$ with $\rho_{\max}$ the maximum link overload ratio under rerouted traffic (soft penalty), and $\chi_{C3}(\delta) \in \{0, 1\}$ indicates routing feasibility for all origin–destination pairs (hard constraint). The final ranking score is:

$$s(\delta) = \phi(\delta) \cdot \max\bigl(0, \hat{p}_{\text{base}} - \hat{p}_\delta\bigr). \tag{4}$$

Any intervention with $\phi(\delta) = 0$ receives score zero and is ranked below all feasible candidates regardless of its predicted probability reduction.

Pairwise fallback. When no single-edge restoration satisfies all three constraints, TCFA falls back to evaluating all pairs (δ_1, δ_2) with $\delta_1, \delta_2 \in \mathcal{E}$, $\delta_1 \neq \delta_2$, applying the same constraint checks and scoring jointly.

Separation of constraint verification from model querying. Rerouted traffic $\tilde{\mathbf{X}}(\mathcal{G}')$ is used solely for constraint checking (C2, C3) and is *not* passed to the GNN classifier. The classifier operates on updated topological features (degree, betweenness centrality) recomputed on $\mathcal{G}'$. This avoids distribution shift: injecting rerouted traffic derived from a different topology configuration would violate the training distribution, yielding unreliable predicted probabilities. The acknowledged cost is that the model cannot directly assess the magnitude of traffic changes induced by interventions; see Section 5.3 for discussion.

Multiplicative coupling ensures any intervention with $\phi(\delta) = 0$ receives exactly zero score, preventing infeasible candidates from outranking feasible ones regardless of predicted probability reduction.

4 Results

4.1 Experimental Setup

Evaluation is conducted on two real backbone topologies: Abilene (12 nodes, 15 edges) [7] and GÉANT (23 nodes, 38 edges) [16], using publicly available traffic matrices from the Internet2 and GÉANT archives, respectively. A GRU-based forecaster is trained on normalised per-edge traffic matrices with sequence length $T = 12$. A timestep t is labelled as a failure if the resulting MAPE exceeds:

$$\tau = \max\bigl(\tau_{\min},\ P_{75}\{\epsilon_t : \epsilon_t > 0\}\bigr), \quad \tau_{\min} = 0.45. \tag{5}$$

The GNN classifier is selected from GCN, GAT, and GraphSAGE based on validation F1-score, and trained on the leakage-free node features described in

Section 3. Topology failures are simulated by uniformly removing $\rho = 20\%$ of edges subject to graph connectivity constraints. The classifier is trained on failure seed 42 and evaluated across five held-out failure seeds $\mathcal{S} = \{7, 13, 21, 55, 99\}$, ensuring no overlap between training and evaluation failure configurations. The dataset is split 70%/15%/15% for training, validation, and testing. All methods operate over the full original edge set without access to ground-truth failed-edge labels, reflecting a realistic operator setting.

The Abilene training set contains only 4.3% failure samples (61/1411), while the test set is 80.9% failure samples (245/303). This asymmetry arises because topology failures are simulated only at evaluation time—the training topology is the healthy graph—whereas test evaluation is conducted exclusively on the degraded topology. To mitigate the resulting class imbalance, the training set is augmented with held-in failure snapshots from seed 42, yielding a mixed training set of 1,594 samples. Class weights are applied via inverse-frequency weighting (normal = 0.153, failure = 0.847). For GÉANT, training and test failure rates are more balanced (27.1% and 22.2%, respectively).

4.2 Metric Definitions

Let $\mathcal{E}_f \subset \mathcal{E}$ denote the ground-truth set of failed edges and $\hat{\pi} = (\hat{e}_1, \ldots, \hat{e}_{|\mathcal{E}|})$ the ranked list of candidate interventions produced by a method.

$$\text{Hit@1} = \frac{1}{N} \sum_i \mathbf{1}\left[\hat{e}_1^{(i)} \in \mathcal{E}_f^{(i)}\right] \tag{6}$$

$$\text{MRR} = \frac{1}{N} \sum_i \frac{1}{\text{rank}_i}, \quad \text{rank}_i = \min\left\{r : \hat{e}_r^{(i)} \in \mathcal{E}_f^{(i)}\right\} \tag{7}$$

Definition 2 (Fix@k). *Fix@k measures the fraction of predicted failures resolved by applying the top-k ranked interventions. Let $\mathcal{G}^{(k)}$ denote the graph after sequentially applying the top-k interventions.*

$$\text{Fix@}k = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}\left[\hat{P}\left(\text{fail} \mid \mathcal{G}^{(k,i)}\right) \leq (1 - \delta_{\text{abs}})\, \hat{P}\left(\text{fail} \mid \mathcal{G}^{(0,i)}\right)\right], \quad \delta_{\text{abs}} = 0.05 \tag{8}$$

A *model-relative* variant Fix@k_{rel} declares success when $(P_{\text{base}} - P_k)/(P_{\text{base}} - P_{\text{healed}}) \geq \delta_{\text{rel}} = 0.40$, where P_{healed} is the predicted probability on the fully-restored original topology.

4.3 GNN Failure Detection

On Abilene, **GraphSAGE** achieves the strongest validation performance (F1 = 0.921, ROC-AUC = 0.896) and is selected as the classifier for this topology. GraphSAGE also leads on GÉANT (F1 = 0.472, ROC-AUC = 0.728), outperforming GCN (F1 = 0.381) and GAT (F1 = 0.397). The lower GÉANT F1 is unsurprising

Table 1. Hit@1 and MRR at $\rho = 0.20$, averaged over five held-out seeds. All methods operate over the full original edge set without access to ground-truth failed-edge labels.

Dataset	Method	Hit@1	MRR
Abilene	Random (all original edges)	0.225	0.225
	Degree heuristic	0.000	0.333
	Betweenness heuristic	0.000	0.333
	Unconstrained (CF-proxy)	1.000	1.000
	TCFA (ours, C1+C2+C3)	**1.000**	**1.000**
GÉANT	Random (all original edges)	0.000	0.000
	Degree heuristic	0.000	0.056
	Betweenness heuristic	0.000	0.200
	Unconstrained (CF-proxy)	1.000	1.000
	TCFA (ours, C1+C2+C3)	**1.000**	**1.000**

given the conditions: traffic is extremely sparse (mean ≈0.01 Mbps), the feature space is larger (10-dimensional vs. 5-dimensional on Abilene), and the failure class makes up only 22.2% of the test set.

4.4 Failure Attribution: Hit@1 and MRR

Table 1 reports Hit@1 and MRR for all methods at $\rho = 0.20$, averaged over five held-out seeds. TCFA achieves Hit@1 = 1.000 and MRR = 1.000 on both topologies. The unconstrained ablation (CF-GNNExplainer proxy) matches these scores, which is expected—model-based scoring alone is sufficient to identify causally implicated edges. What it cannot do, as the Fix@k results show, is translate that identification into interventions that are physically actionable.

4.5 Per-Seed Breakdown

Tables 2 and 3 report Hit@1, MRR, gradient-CFA failure probability reduction, and constraint violation rate individually for each held-out seed. TCFA achieves Hit@1 = 100% and MRR = 1.000 on every individual seed for both datasets. The per-seed constraint violation rates on Abilene vary substantially (6.7%–93.3%), confirming that C2 is active across diverse failure configurations and not just on average.

4.6 Robustness Across Failure Rates

Table 4 reports Hit@1 vs. failure rate ρ on Abilene. TCFA maintains Hit@1 = 100% across all tested rates ($\rho \in \{7\%, 13\%, 20\%, 27\%, 33\%\}$). On GÉANT, TCFA similarly maintains Hit@1 = 100% across all tested failure rates ($\rho \in \{5\%, 8\%, 13\%, 18\%, 25\%\}$). At $\rho = 8\%$, TCFA leads the best heuristic (Betweenness, 33.3%) by +66.7 percentage points; at $\rho = 25\%$ it leads the best heuristic (Random, 40.0%) by +60.0 percentage points.

Table 2. Per-seed results on Abilene ($n = 50$ per seed, $\rho = 0.20$). ViolRate = fraction of unconstrained candidates violating at least one constraint.

Seed	Failed edges	Hit@1	MRR	GradCFA	ViolRate
7	(5,6),(1,11),(4,7)	100%	1.000	99.8%	93.3%
13	(7,9),(5,6),(9,10)	100%	1.000	99.7%	93.3%
21	(1,4),(2,5),(4,6)	100%	1.000	99.9%	87.0%
55	(1,11),(1,5),(4,7)	100%	1.000	99.6%	6.7%
99	(5,6),(7,9),(3,9)	100%	1.000	99.6%	93.3%
Mean		**100%**	**1.000**	**99.7%**	**74.7%**

Table 3. Per-seed results on GÉANT ($n = 50$ per seed, $\rho = 0.20$, 38-edge search space). ViolRate = 0.0% across all seeds reflects GÉANT's sparse traffic regime (mean ≈0.01 Mbps).

Seed	Failed edges	Hit@1	MRR	GradCFA	ViolRate
7	(18,22),(11,20),(12,15)	100%	1.000	100.0%	0.0%
13	(16,21),(17,18),(20,22)	100%	1.000	100.0%	0.0%
21	(6,9),(4,5),(14,22)	100%	1.000	100.0%	0.0%
55	(16,17),(18,22),(14,15)	100%	1.000	100.0%	0.0%
99	(14,22),(18,22),(8,9)	100%	1.000	100.0%	0.0%
Mean		**100%**	**1.000**	**100.0%**	**0.0%**

4.7 Fix@k: Actionability and Constraint Ablation

Table 5 reports Fix@k for $k \in \{1, 2, 3\}$ on both topologies, including per-constraint ablations. The unconstrained ablation ($\phi \equiv 1$) achieves Fix@k = 0% at all k on both datasets. *This is the central result of the paper*: without physical feasibility constraints, model-based scoring alone does not produce actionable interventions, even though it achieves perfect Hit@1.

The per-constraint ablations on Abilene isolate each constraint's contribution. C1-only and C1+C2 both achieve Fix@1 = 10%—higher than TCFA-full at Fix@1 = 0%—because without C3, a single intervention that disrupts a critical routing path can produce a large one-step probability reduction by forcing the model into an unrecognised topology state. However, both partial ablations converge to the same Fix@2 (76%) and Fix@3 (94%) as TCFA-full, confirming that C2 and C3 matter most for sequential multi-step interventions.

4.8 Failure Probability Reduction

On GÉANT, sequential application of the top-3 interventions drives predicted failure probability from $\hat{P}^{(0)} = 1.000$ down to 0.869 at $k = 1$, 0.563 at $k = 2$, before rising slightly to 0.673 at $k = 3$. The non-monotonic behaviour at $k = 3$ occurs because the third intervention triggers a transient C2 violation,

Table 4. Hit@1 vs. failure rate on Abilene ($n = 45$ samples per level, 3 seeds).

ρ	Random	Degree	Betweenness	**TCFA**
7%	8.9%	0.0%	0.0%	**100%**
13%	8.9%	0.0%	0.0%	**100%**
20%	37.8%	33.3%	33.3%	**100%**
27%	40.0%	66.7%	66.7%	**100%**
33%	40.0%	66.7%	66.7%	**100%**

Table 5. Fix@k ablation on Abilene and GÉANT ($\delta_{\text{abs}} = 0.05$ primary; $\delta_{\text{rel}} = 0.40$ model-relative). TCFA Fix@1 = 0% on Abilene reflects the high baseline $\hat{P}^{(0)} = 0.960$: a single intervention yields only a 3.2 percentage-point reduction, below the $\delta_{\text{abs}} = 0.05$ threshold. 95% bootstrap CIs for GÉANT: Fix@1 $\in [70.0\%, 92.0\%]$, Fix@3 $= 100.0\%$.

Dataset	**Method**	**Fix@1**	**Fix@2**	**Fix@3**	**Fix@1$_{\text{rel}}$**	**Fix@2$_{\text{rel}}$**	**Fix@3$_{\text{rel}}$**
Abilene	**TCFA full (C1+C2+C3)**	0%	**76%**	**94%**	0%	**100%**	**100%**
	Ablation: C1 only	10%	76%	94%	0%	100%	100%
	Ablation: C1+C2	10%	76%	94%	0%	100%	100%
	Unconstrained / CF-proxy	0%	0%	0%	0%	0%	0%
	Random baseline	4%	4%	6%	4%	4%	6%
GÉANT	**TCFA full (C1+C2+C3)**	**82%**	**100%**	**100%**	36%	**100%**	**100%**
	Unconstrained / CF-proxy	0%	0%	0%	0%	0%	0%
	Random baseline	10%	10%	12%	10%	10%	12%

temporarily pushing rerouted traffic beyond capacity headroom. Fix@3 = 100% nonetheless holds because the probability at $k = 3$ (0.673) still satisfies the $\delta_{\text{abs}} = 0.05$ threshold relative to the baseline $\hat{P}^{(0)} = 1.000$, since Fix@k measures reduction relative to baseline, not relative to the previous step.

4.9 Capacity Headroom Sensitivity

Table 6 reports Hit@1, MRR, Fix@1, and constraint violation rate across headroom values on Abilene. Hit@1 = 100% and MRR = 1.000 are stable across all tested values, confirming that the attribution ranking is insensitive to the capacity threshold. The constraint violation rate decreases monotonically from 100% (headroom = 1.0×) to 0% (headroom = 3.0×), demonstrating that C2 is an active and discriminating constraint at the paper's chosen value of 1.5×.

4.10 Computational Overhead

Explanation latency is measured over 250 samples on a single GPU (CUDA). On Abilene, mean runtime is 10.69 ± 0.58 s per sample. On GÉANT (38 edges, 10-dimensional features), mean runtime is 86.9 ± 678.9 s per sample; the large standard deviation reflects variable constraint-checking cost across failure configurations that differ in how many edges have simultaneously failed. Both fall

Table 6. Capacity headroom sensitivity on Abilene ($n = 30$ samples, seed 42). Hit@1 and MRR are stable across all headroom values; Fix@1 = 0% reflects the single-step probability reduction (1.9 pp) falling below the $\delta_{\text{abs}} = 0.05$ threshold. Fix@3 = 94% is the paper's primary Fix@k result.

Headroom	Hit@1	MRR	Fix@1	ViolRate
1.00×	100%	1.000	0%	93.3%
1.25×	100%	1.000	0%	93.3%
1.50× (paper)	100%	1.000	0%	74.7%
2.00×	100%	1.000	0%	6.7%
3.00×	100%	1.000	0%	0.0%

within operational constraints for post-alarm root cause analysis in backbone networks, where response cycles typically span minutes.

5 Discussion

5.1 The Role of Physical Constraints in Counterfactual Attribution

The central finding is that topology constraints are not merely a post-hoc correctness filter but a *primary driver* of attribution performance. The Fix@k results make this precise: the unconstrained ablation achieves perfect Hit@1 and MRR—it correctly identifies which edges are causally implicated—yet achieves Fix@k = 0% at all k. TCFA's multiplicative coupling ensures that only interventions whose full feasibility is confirmed are elevated in the ranking, directly translating attribution accuracy into actionability.

The per-constraint ablation in Table 5 clarifies each constraint's contribution. C1-only and C1+C2 both match TCFA-full at Fix@3 = 94% on Abilene, but diverge at Fix@1: partial ablations achieve 10% versus 0% for TCFA-full, because without C3 a single intervention can artificially depress predicted failure probability by forcing the model into an unrecognised topology state.

5.2 Actionability as a First-Class Evaluation Criterion

Fix@k captures a fundamentally different notion of performance compared to Hit@1 or MRR. While Hit@1 evaluates whether the correct edge is ranked first, Fix@k evaluates whether applying the top-k interventions *resolves the predicted failure*—an operationally meaningful criterion that existing GNN explanation benchmarks do not measure. The gap between Hit@1 = 100% and Fix@k = 0% for the unconstrained ablation demonstrates that retrieval accuracy and operational actionability are orthogonal. We argue that Fix@k should accompany Hit@k and MRR as a standard evaluation criterion wherever GNN explanations are intended to guide real-world interventions.

5.3 Limitations and Future Work

Single-path routing assumption. The traffic model assumes single-path shortest-path routing, whereas real backbone networks often employ multi-path strategies such as ECMP or MPLS-TE [1]. Extending constraint verification to multi-path routing would improve realism but increase computational complexity.

Search space scalability. The current search strategy is limited to single-edge and pairwise restorations. Learning-based pruning using the gradient attribution signal to pre-filter candidates could improve scalability to higher-order failure scenarios.

Capacity estimation from historical data. Edge capacities are estimated from historical traffic percentiles rather than operator-defined values. Although sensitivity analysis across headroom values ($1.0\times$–$3.0\times$) confirms Hit@1 = 100% and MRR = 1.000 are stable regardless of threshold, estimates may be less accurate in highly non-stationary traffic environments, and Fix@1 varies with constraint tightness.

Traffic-aware post-intervention prediction. A future traffic-aware variant that recomputes flow features under rerouted demands could improve post-intervention probability estimates, at the cost of potential distribution shift if rerouted traffic falls outside the training regime.

Fix@k inflation under high test failure rates. The model-relative variant Fix@k_{rel} directly addresses this by normalising against the fully-healed baseline probability; Fix@$k_{\text{rel}} = 100\%$ at $k = 2$ on Abilene confirms that inflation does not affect TCFA's results.

Generality across topologies. Evaluation is conducted on two backbone topologies; extending experiments to networks with more heterogeneous degree distributions or dynamic traffic regimes would further validate the generality of the approach.

6 Conclusion

TCFA addresses a core limitation of existing GNN explanation methods: the absence of physical feasibility constraints in the counterfactual search.

The key finding is that the unconstrained ablation achieves perfect Hit@1 yet Fix@k = 0% at all k on both datasets, demonstrating that physical feasibility constraints are the mechanism by which attribution accuracy is converted into actionable operator guidance.

Evaluated on Abilene and GÉANT across five independent held-out failure configurations and failure rates from 7% to 33%, TCFA achieves Fix@3 = 94% on Abilene and Fix@3 = 100% on GÉANT, compared to 6% and 12% for random intervention, while maintaining Hit@1 = 100% and MRR = 1.000 across all evaluated settings. Both multi-path routing support and learning-based candidate pruning are natural next steps, and addressing them would bring TCFA considerably closer to something deployable in a live network environment.

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